

THIRD SEMESTER B. Tech.(ECAM)
MID-SEMESTER EXAMINATION, SEPTEMBER 2024

Course Code- **EAEP0302**
 Course Title- **Signals and Systems**

Max. Marks- 20

Time- 1.5 Hours

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q No.	Questions	Marks	CO
1a ✓	Define different types of signals and systems.	2	CO 1
1b ✓	How an analog signal is converted to a digital signal?	1	
1c ✓	Draw the function $\pi(3t+2)$ when $\pi(t) = 2$: for $t \leq 1/2$, 0 : otherwise	2	
2a ✓	Define following signals-(a) Memoryless (b) time invariant (c) linear (d) stable	2	CO1
2b ✓	Determine whether the following discrete -time signal is periodic determine fundamental period. $X[n] = \sin((6\pi/7)n + 1)$	2	
3a ✓	Determine whether each of the following statements concerning LTI is true or false justify your answers (a) the inverse of a causal LTI system is always a causal. (b) If an LTI system is causal it is stable.	2	CO2
3b ✓	Name some properties of an LTI systems with example.	2	CO2
4a ✓	Express each of the following complex numbers in polar form and plot them in complex plane indicating magnitude and angle (i) $(1+j)e^{j\pi/6}$ (ii) $(1-j\sqrt{3})^3$	1	CO1
4b ✓	The following are the impulse responses of the continuous time LTI systems. Determine whether each system is causal or stable. Justify your answer $h(t) = e^{-4t}u(t-2)$ $h(t) = e^{2t}u(-1-t)$	2	CO2
5b ✓	In $y[n] = x[n]x[n-2]$ (a) Is the system memoryless? (b) determine output when input is $A\delta[n]$. (c) Is the system invertible?	2	CO2
5a ✓	Let $x(t) = u(t-3) - u(t-5)$ and $h(t) = e^{-3t}u(t)$ compute $y(t) = x(t) * h(t)$	2	CO2